CYBER SECURITY CHAT – OCT 2025

TOPICS:

Who am I?

Career Path

Security talk

Investigation

ALEXANDER ROBERTS

Sr. Info Security Specialist

13 years of Experience in Cyber Security

Worked in **Healthcare** for 5 years and **Legal** for 8 years

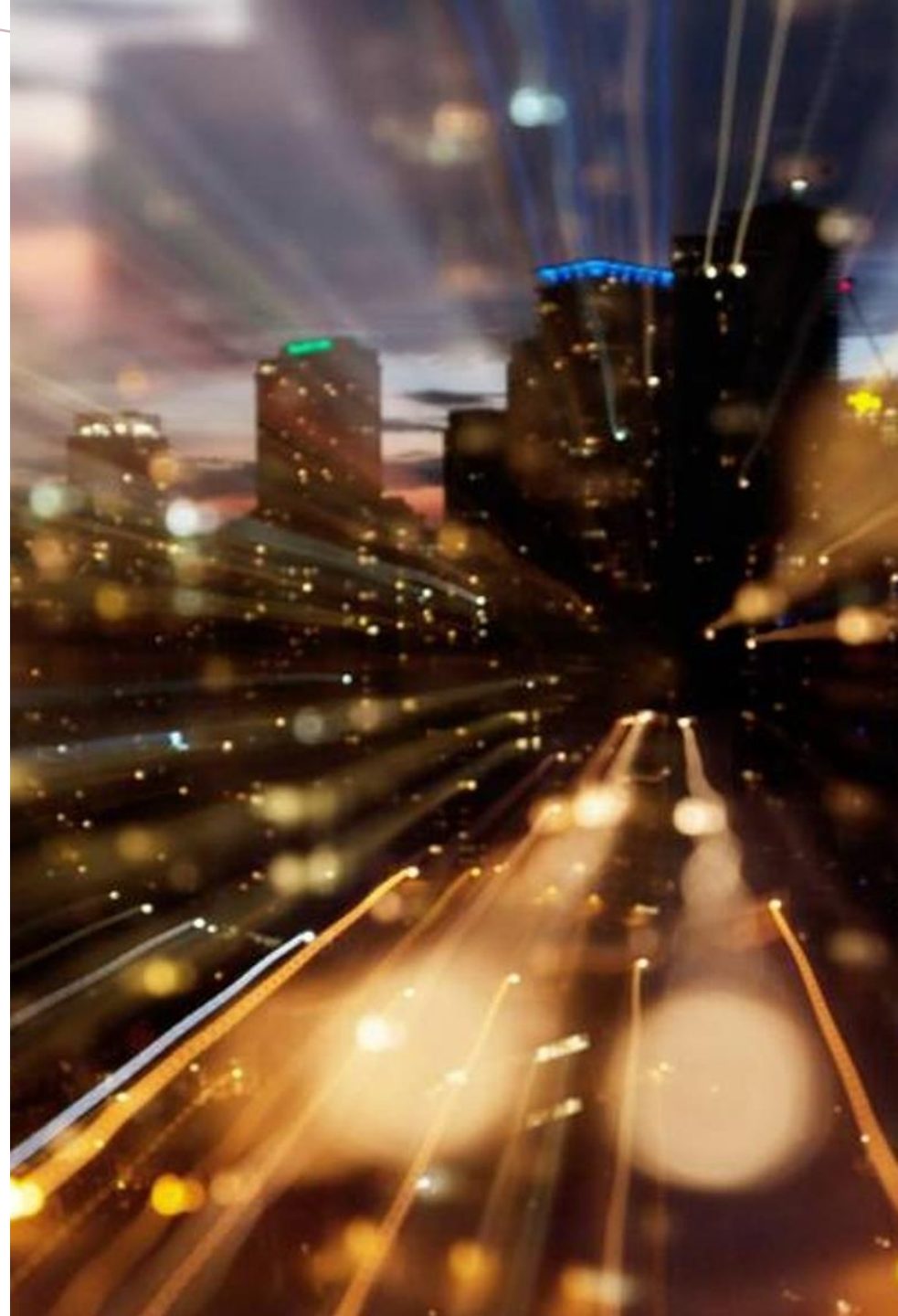
Howard University Alumni

Certifications: CHFI, CEH, CPT, GDAT, GCTI, Security +, ECSS

Random Facts: I practice Shorinji Kempo for over 9 years and I'm a huge Steelers fan.

I DO NOT CODE!!!!!

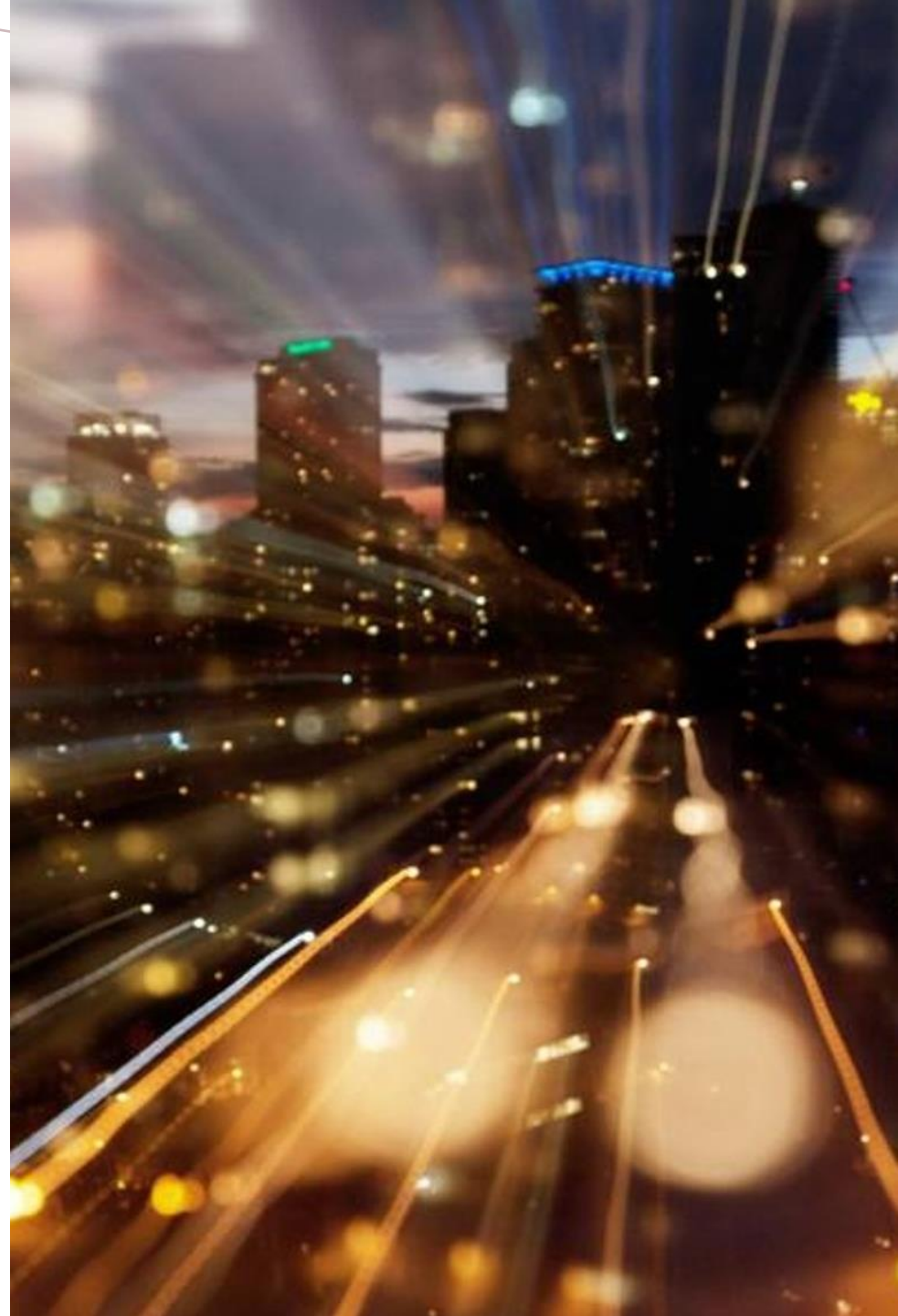
Currently employed at Sidley Austin- SIEM Management, Threat Intel, IR, DLP mgmt, EDR mgmt, SIRT Lead, blah blah blah.



HOW DID I START?

A looooooong time ago back when I was a boy……

- Started with my choice of major. BBA with a Minor in Info Systems and a concentration in Cyber Seuciry
- Networking through internships
- Expressing my interest in a particular fields, which at the time was Incident Response and Forensics.
- Learned from the ground up on to identify malware on a machine without AV. (Reg Keys, Task Manager, CMD line, Event Logs)



HOW DO YOU START?

1. Research security topics/fields
2. Be curious. Ask questions on a particular subject and see if you can answer them.
3. Attend local Security conferences and gatherings
    a. This helps you network with those whom are looking for additional resources
4. Take training(If you can afford it)

The best people in this field understand fundamentally how a computer is functioning. They then take it a step further by trying to either break the system(hackers) or identify parts of the system doing something they shouldn't(Defenders)

It can be rough to identify a starting job in cyber security so starting off in a Server/Network admin, help desk, or even tech support role is how many got into the position they are in.

WHAT IS CYBER SECURITY TODAY?

- Overall all the goal of Cyber Security is to protect the privacy of your actions and data on the internet and your endpoints and ensure uptime.

- We defend from both Internal and External threats. Below are the External adversaries we face.

    - Hacktivist
    - Script Kiddie
    - Black/Grey/White Hat
    - APT – Advanced Persistent Threat (Usually nation sponsored)

- There are a multitude of tools we utilize in the industry such as: Firewalls, Email Filters, AV, EDR, Forensics, Encryption, NDR, RBAC

- What do you think is the weakest part in a security system?

EXTERNAL THREATS

Hackers performing Active or Passive attacks

- Active attacks are live, consistent attacks used to break systems in a short timeframe
- Passive attacks are automated attacks that span a long period of time. (Some for years!)

Attack Methods

- Malware- Trojan, Worm, Timebomb
- Man-in-the-middle Attack
- Spoofing
- X-site Scripting
- SQL Injection
- DoS/DDoS

INTERNAL THREATS

Internal threats come from your own resources!

Attacks Methods

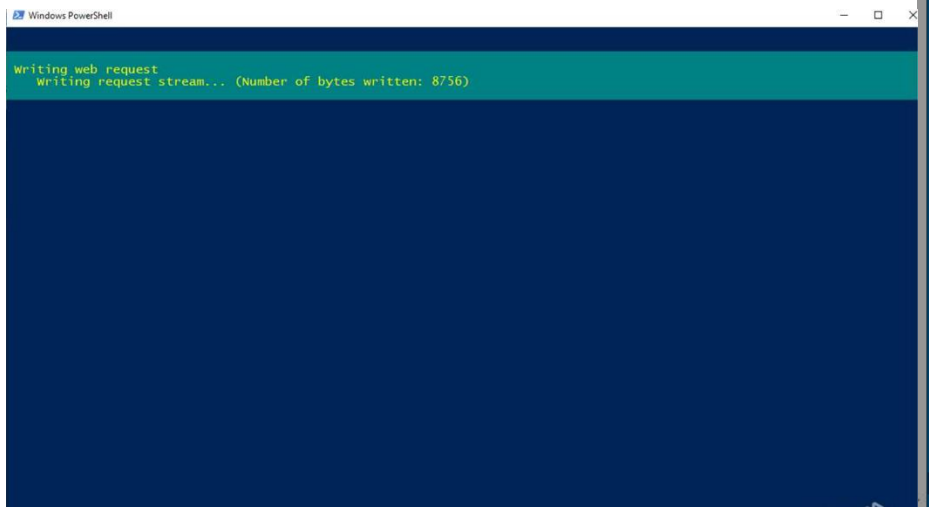
- Social Engineering
- DoS
- Disgruntled employees

EXAMPLE INVESTIGATION

Spear Phishing

Targeted attack on a group of individuals utilizing social engineering to achieve their goal. - INC1069

Malware - https://app.any.run/tasks/9e5a3893-0c6d-4101-9e6b-335c00dfb4c0



NOTES/APPENDIX

Certification roadmap/list - https://pauljerimy.com/security-certification-roadmap

Malware Library/sandbox – App.any.run

Testing site for learning - Hackthissite.org

Free learning resources - Cybrary.it

THANK YOU